# Engineering Portfolio

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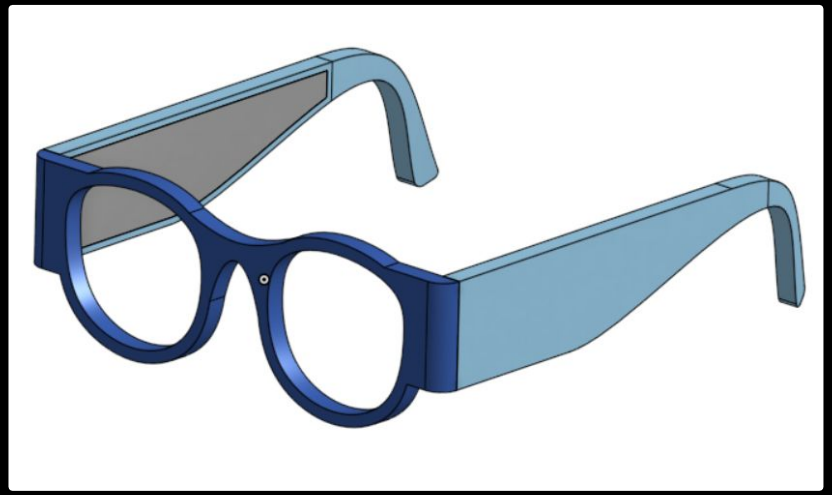
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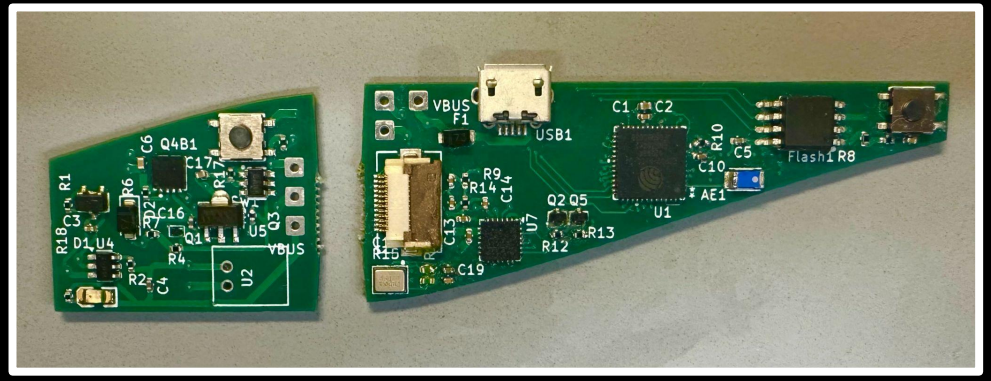
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# AR Speech-to-Text Glasses

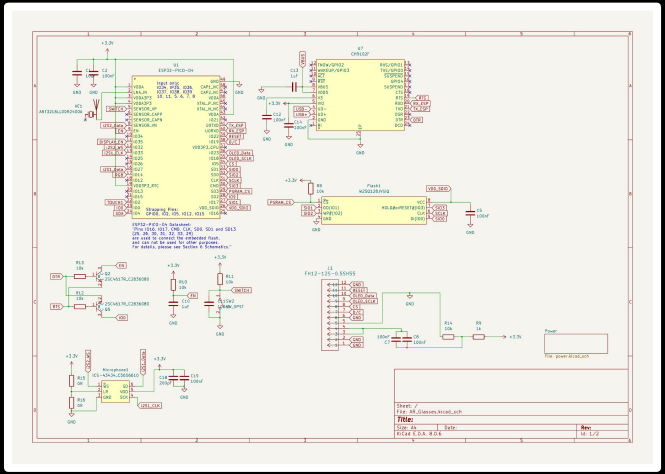
Designing glasses that will display subtitles on the lens by listening to conversations, sending the audio data via bluetooth to a phone or laptop, and sending the analyzed text back to the glasses to display.

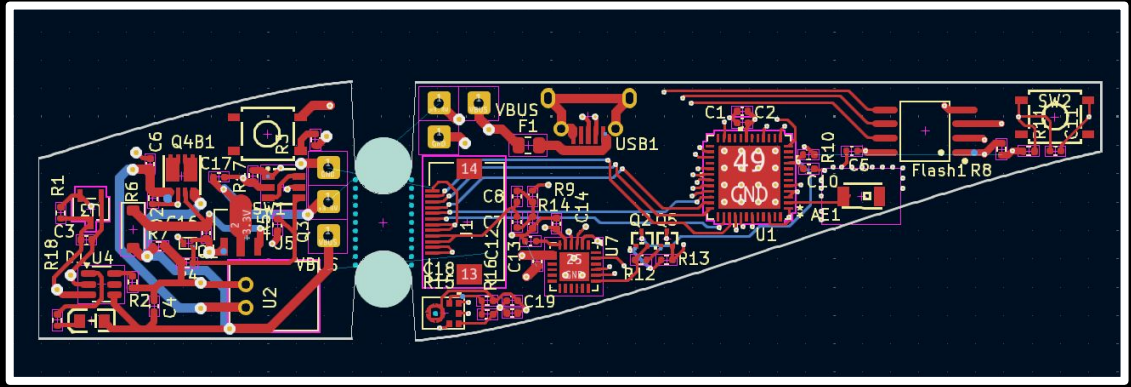


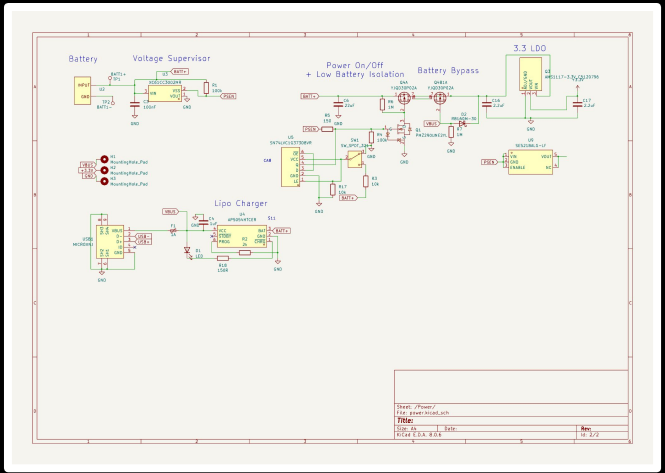



Cad model of the custom glasses frame. Has side panels and a channel in the frame for 3 wires to pass from either side. Used Onshape, but also versed in Solidworks

Two custom PCBs hand assembled at UT Austin's makerspace where I am the Electronics Lead. Left PCB is for power and the right PCB is for processing and display.






The Schematics and PCB Layout of the PCB. Created the entire PCB in around 2 months. Used many online sources and modified an Open Source Smart Glasses model to fit my needs.

Personal Project – Ongoing

# Obstacles

- Unable to have consistent power delivery through PCBs

*Solution*

Split large PCB into multiple smaller, simpler ones for troubleshooting and probing. Still ongoing, but new prototypes have tackled some power issues

Research Assistant – Completed

# Nanoparticle Indentation

Conducted simulations using LAMMPS of high-strain-rate, Aluminum nanoparticle collisions to determine optimal impact velocity for applications such as micro-cold spray.

Publication: Simulation of High Strain Rate Contact of Single Crystal Al Spheres

# Coding for the Simulations

```
#########  NANOPARTICLE VARIABLES   ##################
variable vNP equal -1
variable revvNP equal 1
```

```
#------------------ CREATE LATTICE, ATOMS, AND GROUPS (1) --------------
lattice fcc ${latparam} orient x 1 0 0 orient y 0 1 -1 orient z 0 1 1
# lattice fcc ${latparam} orient x 1 0 0 orient y 0 1 0 orient z 0 0 1
# lattice fcc ${latparam} orient x -1 -1 2 orient y 1 -1 0 orient z 1 1 1
```

```
fix stat all ave/atom 5 20 100 c_strat[*]    # average over 1 ps, 20 samples


dump 2 all cfg 200 /work2/02968/mfbecker/ls6/Modulus_tests/dump.Al_6v1_plate
     xs ys zs id c_peat c_keat f_stat[*] c_volat[1]
dump_modify 2 element Al Al

thermo 10

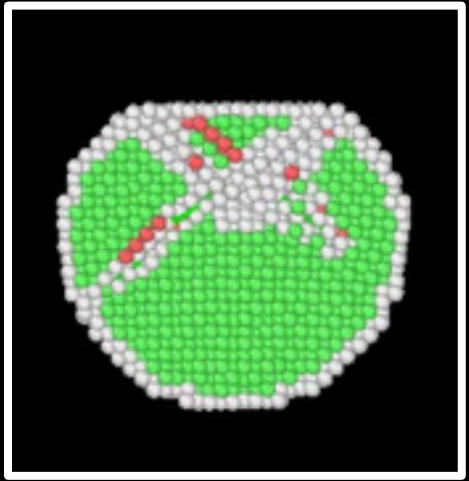
run 10000

# Reverse direction and withdraw indenter
unfix 4
variable rz equal v_z+step*dt*${revvNP}
fix 4 all indent 10.0 plane z v_rz hi units box
fix_modify 4 energy yes

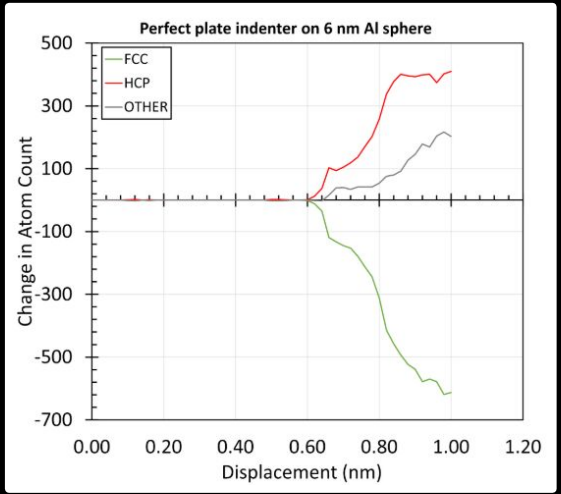
run 2000

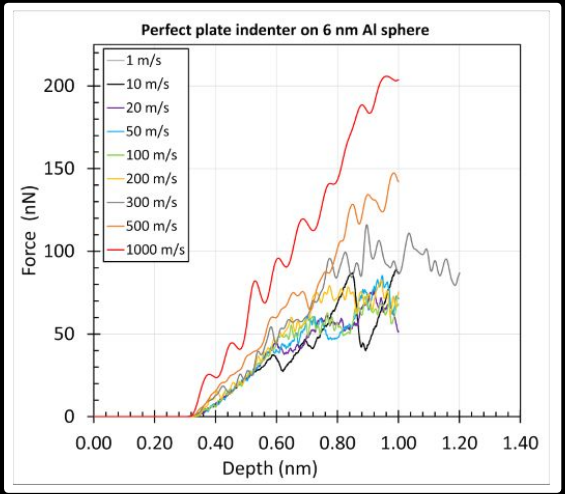
# SIMULATION DONE
print "All done"
```

- The speed of collision and “bounce” back
- Orienting the crystal lattice
- Gathering the data to analyze (timesteps, indent depth, etc.)



Visualization of the data using Ovito software




## Converting Raw Data into graphs

Left: change in normal structure, dislocations and amorphizations

Right: Change in depth of particle with varying speed

Research Assistant – Completed

# Obstacles

- Determining optimal impact velocity
- Understanding crystal terminology and properties

*Solution*

Not being afraid of asking the “wrong” things. Continuously asked questions regarding structural properties of materials. Used knowledge to change variables and achieve ideal deformation.

Personal Project – Completed

# Hand-Tracking Plane

Constructed an RC Plane from scratch that is able to be controlled from tracking my hand movements using my PC's webcam and sending the packets via radio to control the plane

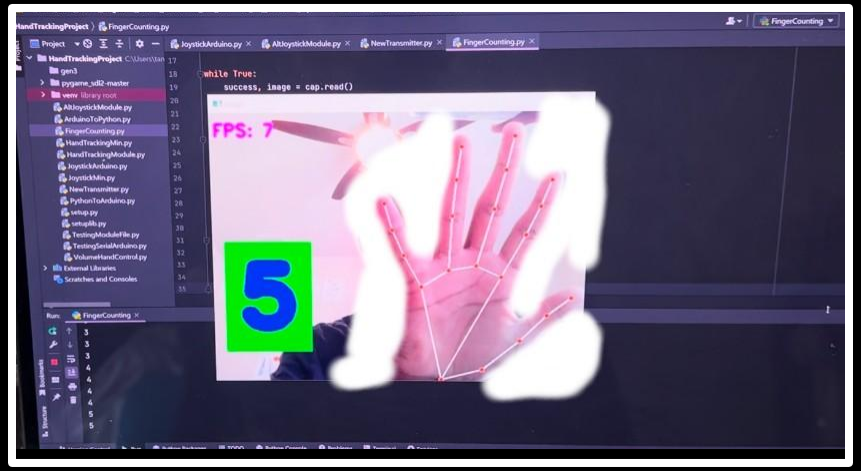


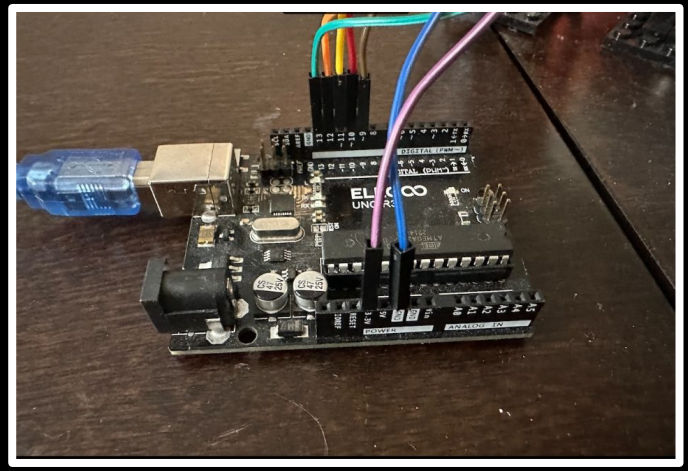


Cut out foam board for the physical body of the plane

---

3D Printed the propellers at the University of Texas at Austin Makerspace where I am the Electronics Lead



The hand-tracking elemented was coded on Python using OpenCV to track the joints of my fingers. I used the change in movement of the joints to control speed and direction.

Used Arduino to parse the Python code and send radio packets to another Arduino on the plane to control servos

Personal Project - Completed

# Obstacles

- Finding optimal motor power to propellor size to aircraft weight ratios

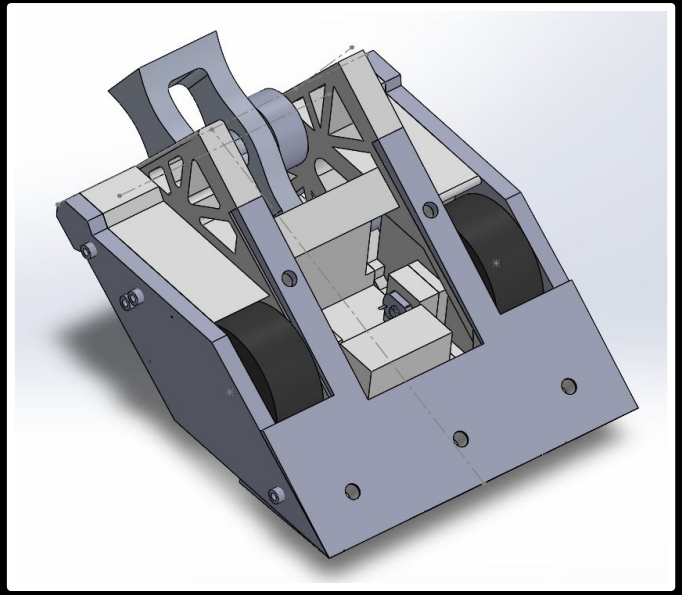
*Solution*

While I used a lot of trial and error, I should have used calculations to have a clearer understanding of how the plane is generating lift.

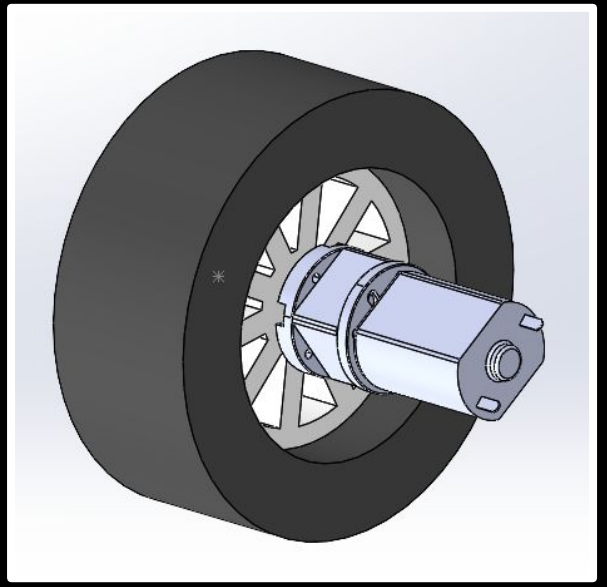
Team Competition – Completed

# Battlebot

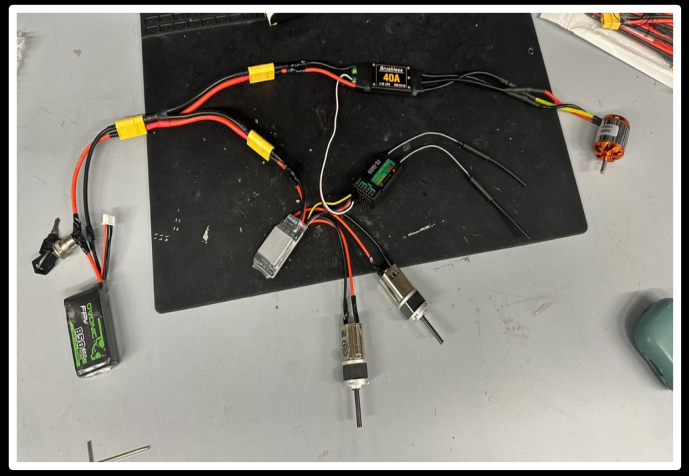
Constructed a robot that would destroy other robots using quick movements, steel spinning weapon, and hard 3D-printed exterior.



Overall CAD Design of Battlebot - Incorporates Weapon, PTEG & TPU Chassis, Drivetrain, Electronics



Drivetrain - In future design, will use a pulley rather than a direct motor to wheel hub




Electronic Circuitry of the Battlebot:

Consists of Lipo Battery, ESC, Radio Transmitter, Drive Motors, Weapon Motor



Partially-Assembled 3D Printed Chassis

Designed to be effective against other bots by adding a ramped surface, trusses, fillets, and bearings

Team Competition – Completed

# Obstacles

- None of our team had experience with Battlebots
- With school and outside conflicts, I ended up being the only person working on it
- Designing for constraints such as weight and size took much longer than I expected

*Solution*

Create prototype design much earlier and allow time for setbacks and revisions